PART 2 · PRODUCT AVAILABILITY REPLY AGENT

Answering from the Data

Every “where can I buy this?” comment gets a reply drafted from the sales reps' shelf visits, and a person sends it in one click.

Ready | Needs a choice | Needs a person | No reply needed

Omar Qasem

THE PROBLEM

Every question means a search through a growing sheet

01

Slow

Someone opens the sheet and filters it by hand for every question, and the sheet keeps growing.

Fix: the reply is drafted before anyone opens the comment.

02

Hard to read

Comments come in Jordanian Arabic or English, with typos, and “it” usually means the product in the post.

Fix: the AI reads the comment, and search links each word to a product, shop or area.

03

Risky to get wrong

A public reply that names the wrong shop, or stock from weeks ago, costs trust.

Fix: shops and dates come only from the data, and every reply says when the shelf was checked.

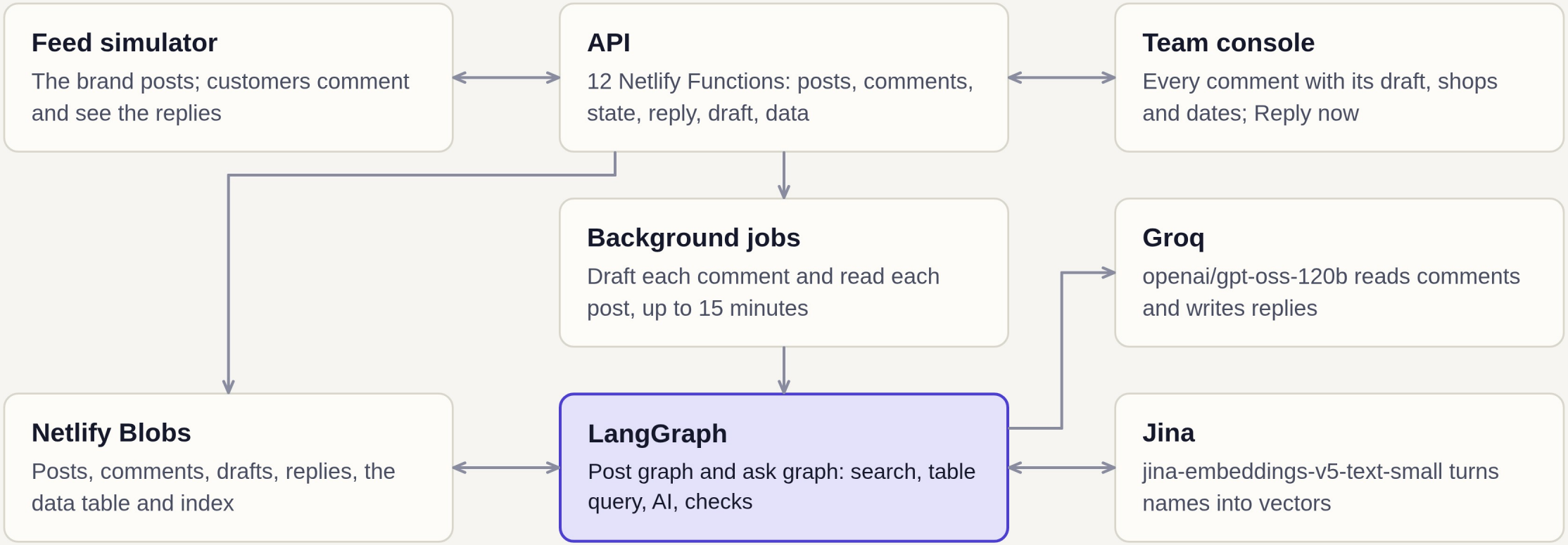
WHAT WOULD YOU BUILD SO THE TEAM ANSWERS IN SECONDS?

A console where every comment already has a checked reply waiting.

Search finds what the customer means. The data table gives the answer. The AI writes it in their dialect, and a person clicks Reply now.

WHAT I BUILT

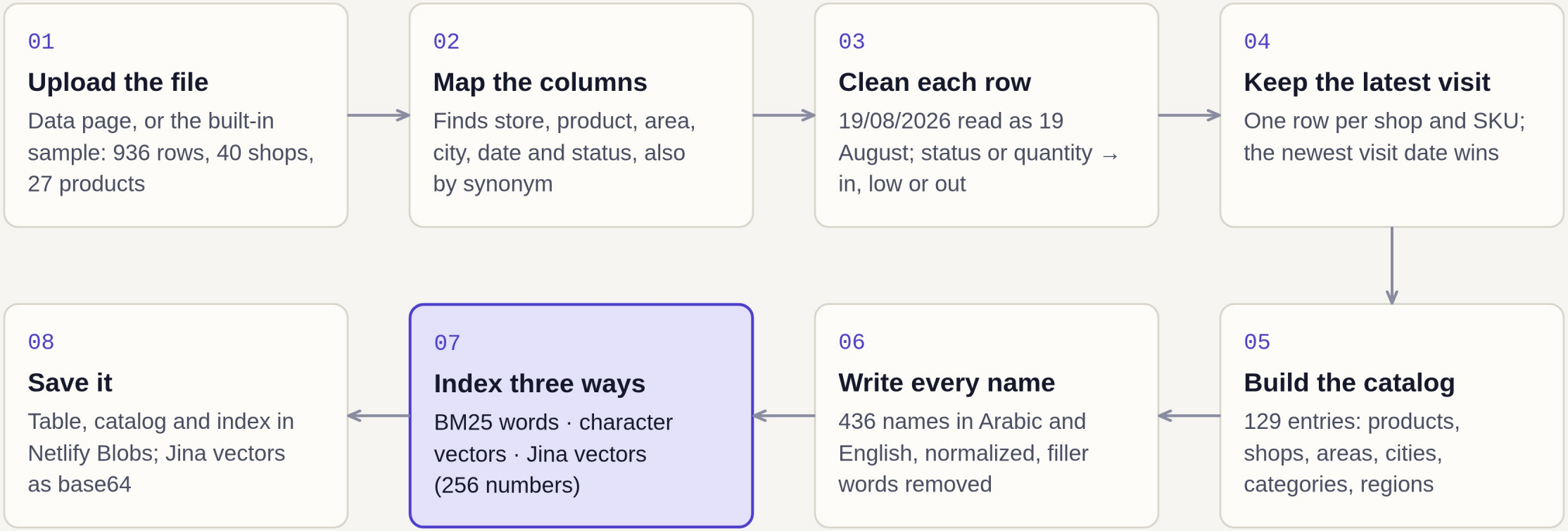
Two small sites and one backend, on Netlify



PATH 1 OF 3 · THE DATA GOES IN

Open it in the explorer ↗

From a CSV to a searchable index in eight steps



Runs on every upload and on “Rebuild index”. A Jina key added later rebuilds the vectors on its own.

Every name is indexed three ways

ONE CATALOG ENTRY

product:olive oil extra virgin

Olive Oil Extra Virgin

زيت زيتون بكر ممتاز

ITS 7 NAMES

زيت زيتون | زيت الزيتون | زيت زيتون بكر | زيت زيتون بكر ممتاز | Olive Oil Extra Virgin | olive oil | extra virgin olive oil

Pack 750 ml · SKU-1013 · Cooking Oil

1 · BM25 on the words

exact words

Each word and its Arabic prefix forms; a word found in fewer names counts more

زيت | زيتون | also matches | زيتون → بزيتون | زيت → والزيت

2 · Character vector

spelling · 18 of 512 slots

#زيت# #زيتون# → pieces of 2, 3 and 4 letters → each hashed to one of 512 slots



3 · Jina vector

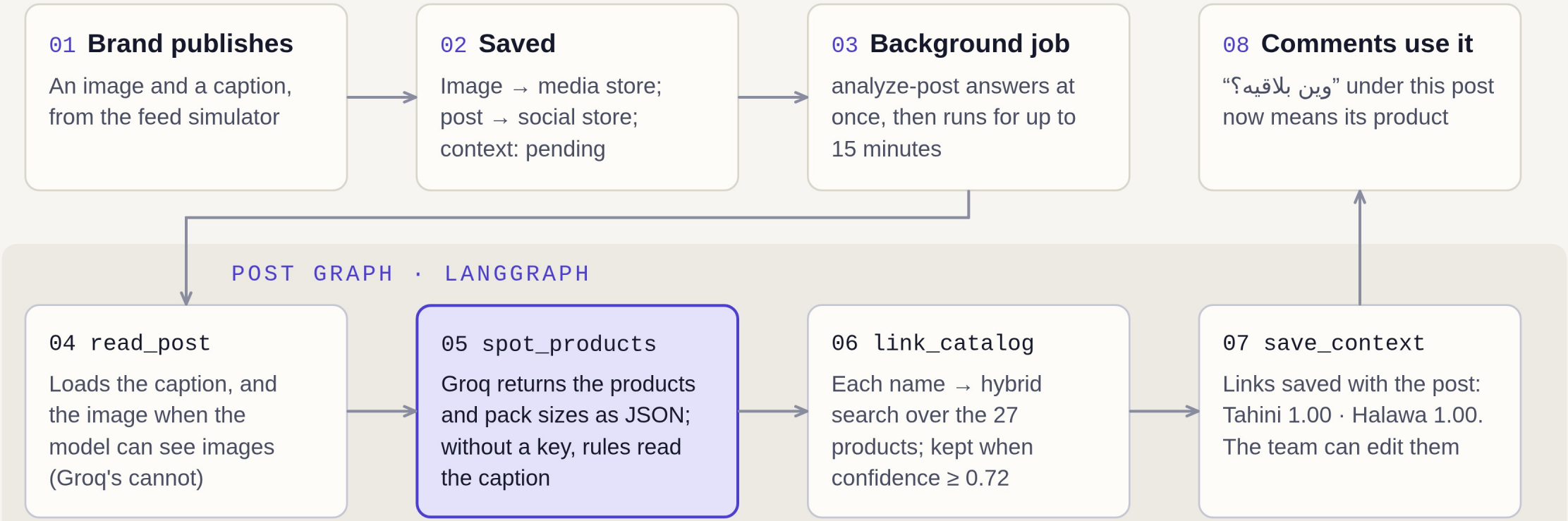
meaning · 256 numbers · schematic

jina-embeddings-v5-text-small gives 1,024 numbers; the first 256 are kept



Open it in the explorer ↗

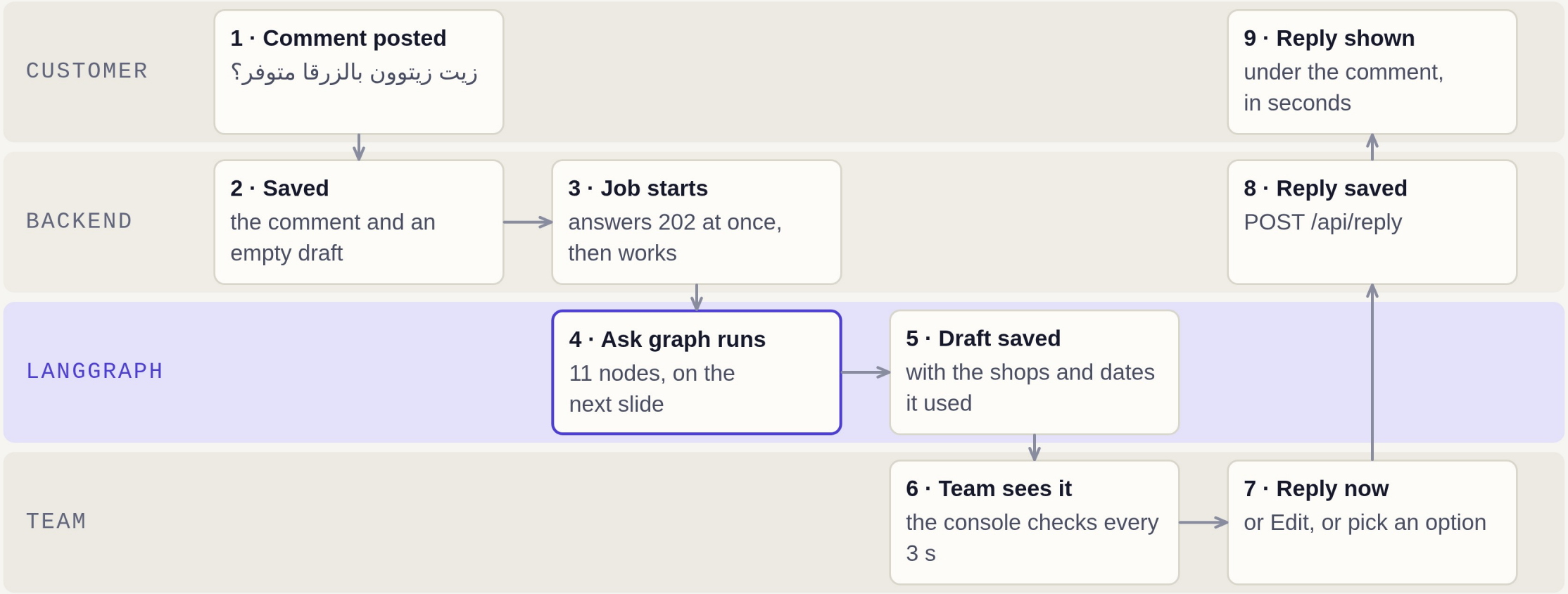
Each post is read once, so “it” has a meaning



The post itself is not added to the index: its products are linked to catalog entries, and the link is stored with the post.

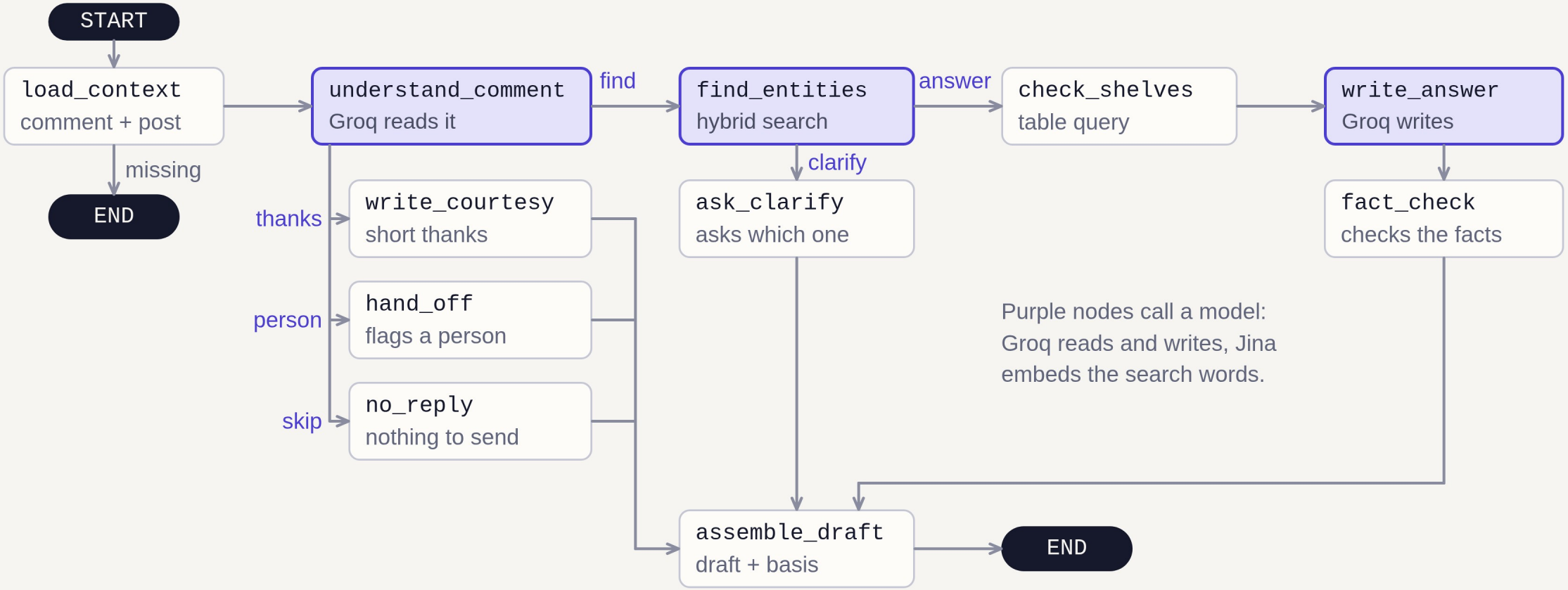
Open it in the explorer ↗

From a comment to a reply on the feed



Click through it in the explorer ↗

Eleven nodes and three decisions



LANGGRAPH · THE ASK GRAPH

The main path, node by node

1 · load_context — no model

Loads the comment, its post and the products linked to the post; waits briefly if the post is still being read.

writes comment · post · post_ctx

2 · understand_comment — Groq

Reads the comment with a strict JSON schema: intent, products, places, shops, size, price. Rules read it if the call fails.

writes understanding

3 · find_entities — Jina + BM25

Links every product and place word to the catalog; a category takes the post's product; unsure means options.

writes products · place · clarify

4 · check_shelves — no model

Filters the table for each product in the place asked, widening area → city → region → country if none is on the shelf.

writes answers

5 · write_answer — Groq

Writes the reply from a JSON of facts only: shops, visit dates, prices if asked; in Jordanian Arabic or English.

writes reply · written_by · model

6 · fact_check — no model

Every shop, date and price must be in the facts, and a visit date must be given; if not, the template replaces the reply.

writes checks · reply

LANGGRAPH · THE ASK GRAPH

The branches, the edges and the shared state

The other five nodes

`ask_clarify`

Asks which product or place was meant; the team gets the options

`write_courtesy`

A short thanks; the model's own words only if they hold no numbers

`hand_off`

A holding reply; the comment is flagged “Needs a person”

`no_reply`

Tags, emojis and spam: nothing to send

`assemble_draft`

One draft: text, status, products, place, shops and dates used, trace

The three decisions

`after load_context`

Comment found → understand_comment; missing → END

`after understand_comment`

Where to buy or price → find_entities · praise → write_courtesy · complaint or other question → hand_off · tag, chat or spam → no_reply

`after find_entities`

A choice is needed → ask_clarify; otherwise → check_shelves

`nine fixed edges`

START → load_context; the answer chain; every branch → assemble_draft → END

THE SHARED STATE

20 channels in one state; each node returns only what it changes; notes and trace append, the rest keep the last value.

SEARCH · INSIDE FIND_ENTITIES

Try any words in the explorer ↗

Hybrid search: three rankings fused into one

1 · BM25 on words

Catches exact names, rare words and Arabic prefixes: بالزرقا → الزرقا

```
Σ idf · 2.2 / (1 + 1.2 · (0.25
+ 0.75 · len/avg))
```

بالزرقا → الزرقاء: 10.30, rank 1

2 · Character vectors

Catches typos and spellings: زيتوون is close to زيتون

```
cos(q, d) = q · d over
512 slots
```

زيت زيتون vs زيت زيتوون: 0.81, rank 1

3 · Jina vectors

Catches meaning, across languages: olive oil is close to زيت زيتون

```
cos(q, d) = q · d over
256 numbers
```

The question is embedded once per search

FUSION · RECIPROCAL RANK FUSION

```
score = 1/(60 + BM25 rank) + 1/(60 + character rank) + 1/(60 + Jina rank)
```

Ranks, not raw scores, so BM25 points and cosines never need a common scale. An exact name adds 0.05.

`match`	confidence ≥ 0.72, no close rival
`ask`	0.55 to 0.72, or a close rival
`none`	below 0.55: no match

Try any words in the explorer ↗

How a vector is computed and compared

Character vector · computed in code

1. Words, with ال removed: زيت · زيتوون
2. Pieces of 2, 3 and 4 letters: #زي · زيت · يتو · توو · ... ز · زي · يت · ت#
3. Each piece hashed (FNV-1a) into 1 of 512 slots
4. Divided by its length, so the vector has length 1

Jina vector · computed by the model

1. The words go to jina-embeddings-v5-text-small
2. 1,024 numbers come back; the first 256 are kept
3. Divided by its length; names once, questions per search

```
nearest = highest cos(q, d) = Σ qᵢ · dᵢ
```

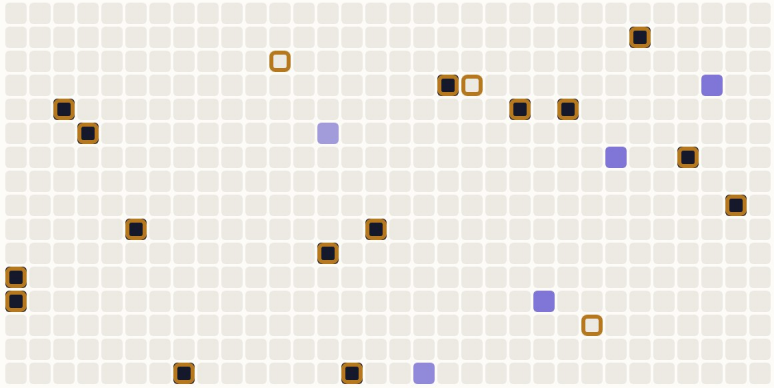
Across all 436 names, in a few milliseconds: no separate vector database is needed at this size.

Character vectors, computed live in this frame

question زيت زيتوون

name زيت زيتون

■ question pieces
□ name pieces
■ shared



cos = 0.81

زيت زيتون	0.81
زيت	0.77
زيت ذرة	0.55
olive oil	0.00

SEARCH · A WORKED EXAMPLE

One comment, with the real numbers

زيت زيتوون بالزرقا متوفر؟ Ahmad, under the olive oil post. “زيتوون” has a typo.

Product words: زيت زيتوون

Name	BM25	Letters	RRF	Conf.
زيت · Cooking Oil	#1	#3	0.0323	0.78
زيت زيتون · Olive Oil	#6	#1	0.0315	0.64
زيت قلي · Sunflower	#2	#6	0.0313	0.52

The category wins, so the post's oil is used: **Olive Oil**. With Groq, the typo is fixed first and “زيت زيتون” is an exact name: 1.00.

Place word: بالزرقا → الزرقا · زرقا

Name	BM25	Letters	RRF	Conf.
الزرقاء · Zarqa city	#1	#1	0.0828	1.00
وسط الزرقاء · area	#3	#3	0.0317	0.71
الزرقاء الجديدة · area	#2	#4	0.0318	0.67

An exact name after normalizing, plus 0.05: a clear **match on Zarqa**, the city.

check_shelves: olive oil in Zarqa, 4 of 5 shops have it on the shelf. The reply names Nour Market Russeifa (24 Aug), Al Amana Grocery Hashemiya (19 Aug) and City Center Zarqa (19 Aug).

Shown: the two rankings that need no API. In production the Jina ranking adds a third term to the same sum.

THE EXACT PART · CHECK_SHELVES

The table gives the answer, not the model

1. Rows for the product, and the pack if one was asked (“2 كيلو”, “the big one”)
2. In the place asked: a shop, an area, a city or a region
3. None on the shelf there? Widen: shop → area → city → region → Jordan
4. Sort: nearest, in stock before low stock, newest visit, more units
5. Up to 3 shops with their visit dates; prices only when asked

No model is involved here, so every shop, status, price and date comes from the data.

EXAMPLE · OLIVE OIL IN ABDOUN

`Abdoun 0 of 1` Sameh Mall Abdoun: out of stock on 24 Aug

`Amman 11 of 14` widened one level up, to the city

Hi Emily! Our reps' last visit (24 Aug) didn't find extra virgin olive oil on the shelf in Abdoun, but you can find it in Amman at:

- Safeway Shmeisani (last visit 22 Aug)
- Al Rawnaq Grocery Jabal Nasr – Jabal Al Nasr (last visit 20 Aug)
- Baraka Mini Market Nazzal (last visit 20 Aug)

Worth checking with the shop before you go

The template reply for this answer (emoji removed)

GUARDRAILS

Five checks before anything is public

01

Facts only

The model writes from a small JSON of facts. It never sees the sheet.

02

Fact check

Every shop, date and price in an AI reply must be in those facts, or the template is used.

03

Unsure? Ask

Below 0.72, or two close names, the draft asks; the team picks with one click.

04

People for people

Complaints and other questions get a holding reply and “Needs a person”.

05

A person sends it

Nothing is posted until someone clicks Reply now. Edit and Dismiss are always there.

If something fails

It still works

No key, an AI error or a daily limit: rules read, templates write. No LangGraph: a built-in runner.

Every draft also shows how it was made: each node, its note and its time, plus the shops and visit dates it used.

THE MODELS

Two AI models, three plain methods

Job	Model or method	Settings	Why
Read comments, write replies	Groq · openai/gpt-oss-120b	Strict JSON schema, low reasoning, up to 1,000 output tokens	Fast, free tier, follows a schema
Search by meaning	Jina · jina-embeddings-v5-text-small	256 numbers, text-matching task	Arabic and English, free key
Search by words	BM25	k1 1.2, b 0.75, Arabic prefix forms	Exact names, rare words
Search by spelling	Character vectors	2–4 letter pieces, 512 slots	Typos; no API needed
Fuse the three	Reciprocal rank fusion	score = Σ 1/(60 + rank)	No score scales to tune
Run the steps	LangGraph.js	2 graphs: post and ask	Explicit nodes, edges, state

gpt-oss-120b: OpenAI's open-weight mixture-of-experts model, 117B parameters with 5.1B active per token. jina-embeddings-v5-text-small: 677M parameters, 119+ languages. Gemini or OpenAI can replace them through one setting.

TESTING

Tested piece by piece, then end to end

42/42 place and product names found the right catalog entry, with no AI

22 sample comments: dialect, typos, sizes, complaints, tags and English

End to end in a browser, from the unzipped deploy file, in five modes: no key, Groq + Jina, OpenAI, Gemini, Jina only

Each AI service replaced by a local stand-in with the same requests, errors and limits

What the tests changed

01 · “رز”

It matched Egyptian rice on its own. Now two names that both hold the word become a choice for the team.

02 · “الحجم الوسط”

“The medium size” was read as the Central region. Size words are now removed before places are matched.

03 · RATE LIMITS

Groq's free 8,000 tokens a minute is too little for 18 drafts at once. Now one job drafts them in turn and waits.

OPTIONS I WEIGHED

Four other ways to build it, and why not

Give the AI the whole sheet

936 rows and growing. It can misread a row or invent a shop, and every question pays for the whole sheet.

Instead: the table query finds the facts; the AI only writes.

Vector search alone

Embeddings blur exact names and short Arabic words with typos, and everything depends on one API.

Instead: three rankings fused; two of them need no API.

Reply automatically

These are public replies in the brand's name. Trust in the drafts has to be earned first.

Instead: drafts in seconds, a person sends; track how often drafts are edited.

n8n again, as in Part 1

This work needs an index, a data table and branching on what the customer meant, not a fixed pipeline.

Instead: LangGraph in code, the option I set aside in Part 1.

KNOWN GAPS

What is simulated, and what comes next

Today	Next step
The feed is a simulator	Meta's webhooks read the real Instagram and Facebook comments, and the Graph API posts the reply
Visits in the sample end on 25 Aug	A daily file from the ERP; the console already warns when the data is over 30 days old
Groq's free tier: 8,000 tokens a minute	Starting drafts take about 10 minutes; a paid tier or a smaller reading model removes the wait
Arabic shop names built from a word list	A one-time review by the team, saved with the data
Index kept as one file in Netlify Blobs	Fine for 436 names; at scale, Postgres with pgvector and the same hybrid search
Quality is not measured yet	Track how often a draft is edited before sending, per type of comment

LIVE DEMO

A comment from the feed to a reply

IN THE DEMO

1. On the feed, comment as a customer under the olive oil post: “وين بلاقيه بإربد؟”
2. The draft appears in the console; open “How this draft was made”
3. Click Reply now; the reply shows under the comment
4. On “وعندكم رز؟”, pick Basmati and get a full answer
5. Publish a post; its products are linked on their own

WATCH FOR

زيت زيتوون بالزرقا متوفر؟ a typo and a prefix	Ready
شاي أخضر ... بالجبيهة؟ none there, so shops in Amman	Ready
الحجم الوسط بكم؟ the medium size, price only	Ready
وعندكم رز؟ وين بلاقيه بماركا؟ Egyptian or Basmati?	Needs a choice
حلاوة ... كانت منتهية a complaint	Needs a person
@sara.haddad هاد اللي حكيتلك عنه a tag	No reply needed

SUMMARY

Answers in seconds, only from the data, sent by a person

Ready before anyone looks

Every comment is read, searched and drafted in the background.

Facts from the table

Search finds the meaning; the shops and dates come from the data and are checked.

A person decides

Reply now, edit, pick an option or hand it to a colleague.